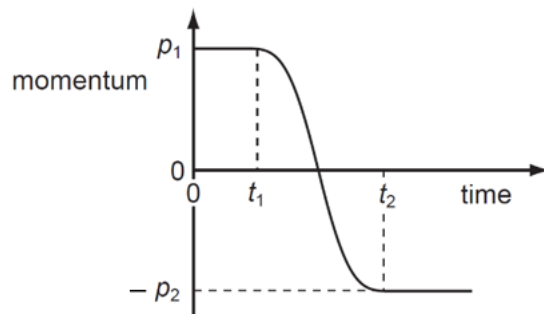


- 3 The graph shows the variation with time of the momentum of a ball as it is kicked in a straight line.



Initially, the momentum is p_1 at time t_1 . At time t_2 the momentum is p_2 .

What is the magnitude of the average force acting on the ball between times t_1 and t_2 ?

A $\frac{p_1 - p_2}{t_2}$

B $\frac{p_1 - p_2}{t_2 - t_1}$

C $\frac{p_1 + p_2}{t_2}$

D $\frac{p_1 + p_2}{t_2 - t_1}$

Ans: D – $\langle F \rangle = \Delta p / \Delta t = p_f - p_i / (t_2 - t_1) = -p_2 - p_1 / (t_2 - t_1) = -(p_2 + p_1) / (t_2 - t_1)$